

Principles Of Cryptography

Endsemester Examination - 2025

- Q) Analyze how the AES key schedule design enhances overall security and ensures effective diffusion throughout the encryption process.

A:

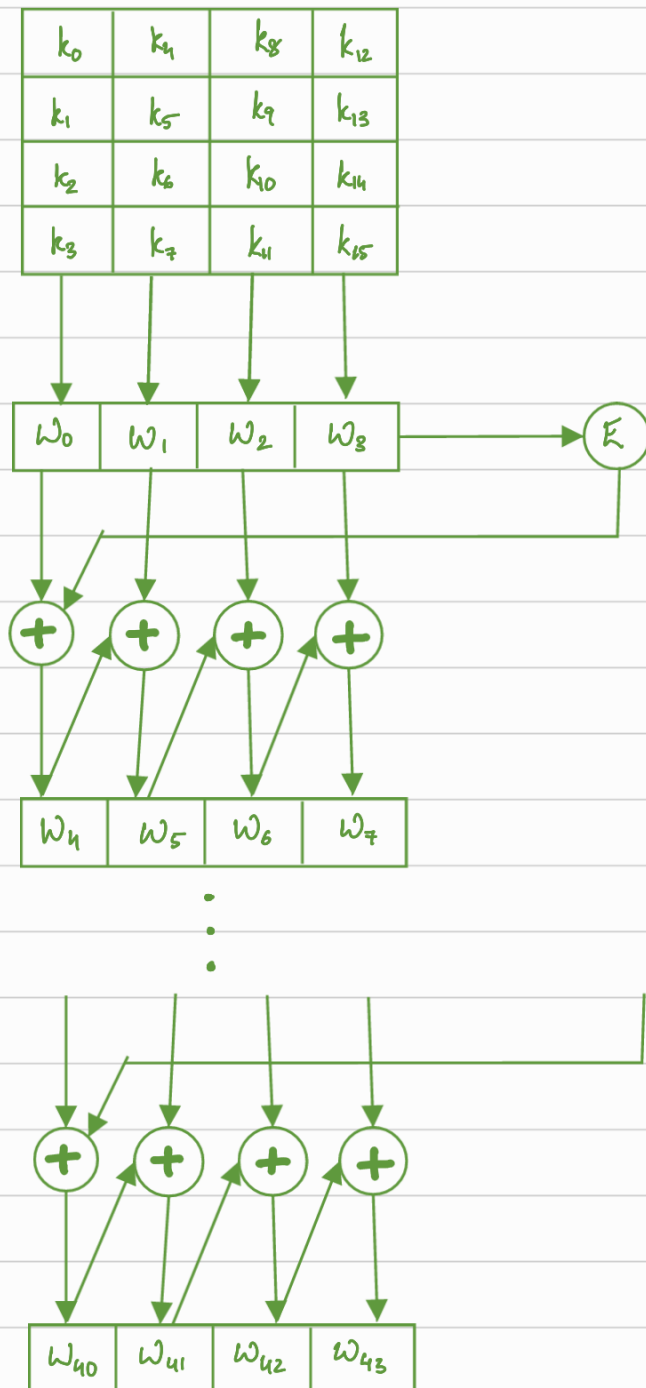


fig:- Overall Algorithm

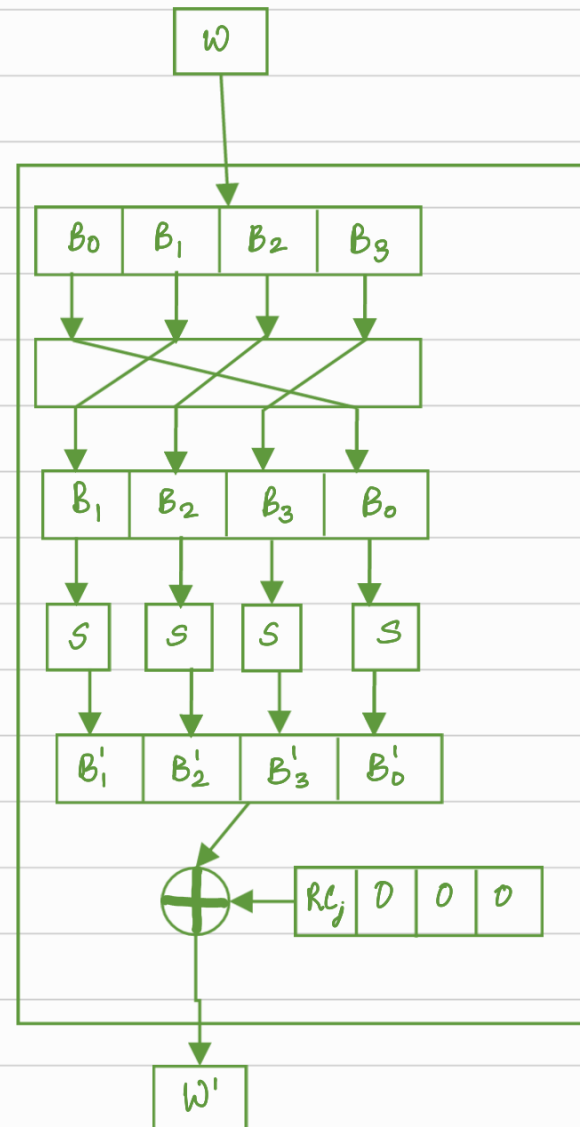


fig:- Function F

- The AES key schedule expands the original cipher key into multiple round keys, ensuring that each encryption round uses a unique key.
- This design enhances security by making the ciphertext highly resistant to cryptanalytic attacks, as even a small change in the key produces completely different round keys, creating a strong avalanche effect.
- It also contributes to diffusion by spreading the influence of the original key across all rounds, ensuring that every part of the ciphertext is affected by the key.
- The use of non-linear transformations and round constants in the key schedule prevents predictable patterns, reinforcing security.
- Overall, the AES key schedule is essential in maintaining both the diffusion and robustness of the algorithm.

Q: Given the plaintext "CRYPTOGRAPHY" and the key matrix:

$$K = \begin{bmatrix} 3 & 4 & 5 \\ 5 & 4 & 2 \\ 2 & 1 & 4 \end{bmatrix}$$

Encrypt the plaintext using the Hill cipher.

A: CRYPTOGRAPHY $\rightarrow \text{len}(p.t) = 12$

$$p.t = \begin{bmatrix} C & R & Y \\ P & T & O \\ G & R & A \\ P & H & Y \end{bmatrix} = \begin{bmatrix} 2 & 17 & 24 \\ 15 & 19 & 14 \\ 6 & 17 & 0 \\ 15 & 7 & 24 \end{bmatrix}$$

a-0	i-8	q-16	y-24
b-1	j-9	r-17	z-25
c-2	k-10	s-18	
d-3	l-11	t-19	
e-4	m-12	u-20	
f-5	n-13	v-21	
g-6	o-14	w-22	
h-7	p-15	x-23	

$$C = \begin{bmatrix} 2 & 17 & 24 \\ 15 & 19 & 14 \\ 6 & 17 & 0 \\ 15 & 7 & 24 \end{bmatrix} * \begin{bmatrix} 3 & 4 & 5 \\ 5 & 4 & 2 \\ 2 & 1 & 4 \end{bmatrix} \pmod{26}$$

$$C = \begin{bmatrix} (2 \times 3 + 17 \times 5 + 24 \times 2) & (2 \times 4 + 17 \times 4 + 24 \times 1) & (2 \times 5 + 17 \times 2 + 24 \times 4) \\ (15 \times 3 + 19 \times 5 + 14 \times 2) & (15 \times 4 + 19 \times 4 + 14 \times 1) & (15 \times 5 + 19 \times 2 + 14 \times 4) \\ (6 \times 3 + 17 \times 5 + 0 \times 2) & (6 \times 4 + 17 \times 4 + 0 \times 1) & (6 \times 5 + 17 \times 2 + 0 \times 4) \\ (15 \times 3 + 7 \times 5 + 24 \times 2) & (15 \times 4 + 7 \times 4 + 24 \times 1) & (15 \times 5 + 7 \times 2 + 24 \times 4) \end{bmatrix}$$

$$C = \begin{bmatrix} 139 & 100 & 140 \\ 168 & 150 & 169 \\ 103 & 92 & 64 \\ 128 & 112 & 185 \end{bmatrix} \pmod{26}$$

$$C = \begin{bmatrix} 9 & 22 & 10 \\ 12 & 20 & 13 \\ 25 & 14 & 12 \\ 24 & 8 & 3 \end{bmatrix} = \begin{bmatrix} J & W & K \\ M & V & N \\ Z & O & M \\ Y & I & D \end{bmatrix}$$

$\Rightarrow JWK M V N Z O M Y I D$

